

# ***Project Report***

## ***Halma-AI***

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Course: Artificial Intelligence

Instructor: Ms. Alishba Subhani

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### **1) Executive Summary**

This project focuses on developing an AI-powered version of the strategic board game Halma. The objective is to implement a competitive AI opponent using the Minimax algorithm with Alpha-Beta Pruning. The AI uses custom heuristics to evaluate board states, considering distance to target, board control, and piece activity. An interactive graphical interface was developed using Pygame, enabling human players to engage in challenging gameplay against the AI.

### **2) Introduction**

#### **Background**

Halma is a strategic board game involving the movement of pieces across the board into the opponent's starting zone. The game was selected for its emphasis on planning and foresight, making it ideal for testing AI strategies. We introduced AI control using Minimax with Alpha-Beta Pruning and heuristic evaluation to enhance gameplay.

#### **Objectives of the project include:**

- Implementing Minimax-based AI with Alpha-Beta Pruning.
- Designing custom heuristics for intelligent decision-making.
- Creating an interactive GUI for player vs AI gameplay.
- Evaluating AI performance at different difficulty levels.

### **3) Game Description**

#### **Original Game Rules:**

Halma is played on a square board where each player moves pieces to the opposite corner. Players can move one step or jump over others.

#### **Innovations and Modifications:**

- Integration of AI with Minimax and Alpha-Beta pruning.
- Heuristic-based decision-making.
- Visual representation of AI thinking.
- Adjustable AI difficulty.

### **4) AI Approach and Methodology**

#### **AI Techniques Used:**

- Minimax Algorithm
- Alpha-Beta Pruning for efficiency.

#### **Heuristic Design:**

- Distance to target zone.
- Board control (blocking opponent).
- Penalize idle pieces.

#### **Complexity Analysis:**

Best Case:  $O(b^{d/2})$ , Worst Case:  $O(b^d)$  depending on pruning. The AI depth is tuned to balance performance and response time.

### **5) Game Mechanics and Rules**

#### **Modified Game Rules:**

- AI plans strategic moves based on heuristics.
- Visualizes move logic to the player.
- Multiple difficulty levels via depth control.

#### **Turn-based Mechanics:**

- Human and AI take turns.
- AI evaluates the best move each turn.
- The game ends when one player reaches the target zone.

#### **Winning Conditions:**

First to occupy the opponent's home zone or block all opponent moves.

### **6) Implementation and Development**

#### **Development Process:**

- Board setup and implementation rules.
- Minimax and Alpha-Beta coding using python language
- Heuristic tuning using mathematical libraries

- GUI development and final integration.

#### **Programming Languages and Tools:**

- Language: Python
- Libraries: Pygame, Copy, math libraries, Numpy
- Tools: GitHub (version control)

#### **Challenges Encountered:**

Balancing AI depth with speed, and visualizing AI decision logic clearly.

### **7) Team Contributions**

Huzaifa Ahmed Khan: AI algorithm development (Minimax, Alpha-Beta Pruning).

Ayesha Ahmed: Game rule modifications and GUI design.

Hafsa Shakeel: Heuristic design and testing.

### **8) Results and Discussion**

The AI demonstrated competitive performance, winning 70% of test matches at depth 3, with an average decision time of 1.5 seconds. Players found the AI challenging and engaging, especially at higher difficulty.

### **9) References**

- Stuart Russell & Peter Norvig, 'Artificial Intelligence: A Modern Approach'
- Research papers on Minimax & Alpha-Beta Pruning
- Pygame Documentation
- Online resources on Halma rules and strategy